On The Cycle Double Cover Problem

Ali Ghassâb[1]

Dedicated to Prof. E.S. Mahmoodian

Abstract

In this paper, for each graph G, a free edge set F is defined. To study the existence of cycle double cover, the naïve cycle double cover of $G \cup F$ have been defined and studied. In the main theorem, the paper, based on the Kuratowski minor properties, presents a condition to guarantee the existence of a naïve cycle double cover for couple (G, F). As a result, the cycle double cover conjecture has been concluded. Moreover, Goddyn's conjecture - asserting if C is a cycle in bridgeless graph G, there is a cycle double cover of G containing C - will have been proved.

[1] Ph.D. student at Sharif University of Technology
e-mail: ghassabali@gmail.com
Faculty of Math, Sharif University of Technology, Tehran, Iran

Cycle Double Cover: History, Trends, Advantages

A cycle double cover of a graph is a collection of its cycles covering each edge of the graph exactly twice.
G. Szekeres in 1973 and, independently, P. Seymour in 1979 conjectured:
Conjecture (cycle double cover). Every bridgeless graph has a cycle double cover.
Yielded next data are just a glimpse review of the history, trend, and advantages of the research. There are three extremely helpful references: F. Jaeger's survey article as the oldest one, and M. Chan's survey article as the newest one. Moreover, C.Q. Zhang's book as a complete reference illustrating the relative problems and rather new researches on the conjecture.
A number of attacks, to prove the conjecture, have been happened. Some of them have built new approaches and trends to study. It will have been pointed some straight attacks, and opposite ones.
Hunting the proof has been caused many researches on a minimal potential counterexample which are named by snarks. In fact, snarks had been born many years before introducing the conjecture. However, nowadays, the importance of studying snarks might be referred to the conjecture.
One by one, the properties have recognized. Easily seen, the minimal potential counterexample must be connected, and also simple. H. Fleischner's vertex-splitting lemma, in 1992, showed the minimal counterexample had to be nonseparable cubic. Furthermore, by applying Vizing's Theorem, it has to have chromatic index four.
Such these results are constructed the definition of snark as a simple connected cubic graph with chromatic index four. The definition has been changed, during the research process. For example, Chan points to a strong version containing cyclically 4-edge-connected property.
Since the finiteness of all snarks might be restricting the conjecture, an approach was to prove its finiteness. However, R. Isaacs, in 1975, introduce an infinite family of snarks.
Elusive to prove the conjecture, L. Goddyn, as a generalization, conjectured if C is a cycle in bridgeless graph G, there is cycle double cover of G containing C. He introduced another trend when showed a minimal potential counterexample must have girth at least ten. A. Huck promoted the result to twelve. He used not only computer programming, but also a method of reducible configuration. That is an approach to catch new results by finding the properties that the minimal potential counterexample has not to have; for instance, triangle-freeness.

In a period of time, it had been conjectured that snarks have bounded girth. However, M. Kochol constructed a snark family with arbitrarily large girth.
Notably, for a special family of graph, the conjecture has been proved. According to a proposition, it was proved that in a nonseparable plane graph $G \ncong K_1, K_2$, each face is bounded by a cycle. Hence, as a consequence, the conjecture will be proved for bridgeless planar graphs.
In another trend, a stronger conjecture is studying to prove the conjecture. For example, the conjecture will be proved if a surface embedding exists such that each face boundary is a cycle. W.T. Tutte, in the mid-1960s, and N. Robertson, in 2007, conjectured for each loopless nonseparable graph there exist such surface embedding.
Some advantages have been found when, with connectivity aspect, the conjecture has been studied. For example, to show that a graph has a cycle double cover, it is sufficient to reduce the problem to each its block.

P.A. Kilpatrick, in 1975, and Jaeger, in 1976, proved that each 4-edge-connected graph has a cycle double cover. Therefore, the minimal potential counterexample has exactly connectivity three.
It has been proved that to show that a 3-connected cubic graph has a cycle double cover, it is sufficient to reduce the problem just for each of its essentially 4-edge-connected components. Hence, the potential counterexample must be essentially 4-edge-connected cubic graph. Moreover, as which has been pointed, it has chromatic index four.

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In this paper, the usual trends, straightly, is not continued. A new approach to study the conjecture will be introduced. In short words, the method is based on the Kuratowski minor properties.

For each graph G, a free edge set F is defiend. Moreover, maximal 2-connectivity components of G are considered, then just special (G, F) are focused on. An algorithm is introduced to use these couple (G, F). The algorithm, based on the behavior of the $cone$ operator and subgraphs which are isomorphic to subdivision of Kuratowski minors, helps to prove of a generalized form of the cycle double conjecture. As an application of the main introduced theorem, which is stronger than the cycle double problem, Goddyn's conjecture will be proved.

Basics

To review all (usual and of course not new) definitions, reference [1] is suitable. However, some usual definitions are reviewed.

Notation. Let G be a graph. As usual, $V(G)$ and $E(G)$ denotes the set of vertices and edges of G in sequential order.

Notation. If $A \subseteq E(G)$, $< A >$ denotes a graph induced by edge set A. For example, if $e \in E(G)$, and its ends are two vertices u and v, then $V(< e >) = \{u, v\}$.

Notation. If $B \subseteq V(G)$, $< B >$ denotes a graph induced by vertex set B. Moreover, $G - B$ means $< V(G) - B >$.

Notation., then means the edge cut between two components $< X >$ and $< Y >$.

Definition. If $X, Y \subseteq V(G)$, define $E(X, Y) \coloneqq \{uv \in E(G) | u \in X, v \in Y\}$. If $Y = V(G) - X$, $E(X, Y)$, denoted by $\partial(X)$, is called an edge cut set in G respect to X.

Definition. e is called a cut edge in G provided there exists $X \subseteq V(G)$ such that $\{e\} = \partial(X)$. Moreover, sometimes the word "bridgeless" is used instead of the expression "without cut edge".

Notation. A sequence $v_1 e_1 v_2 e_2 v_3 \dots e_{k-1} v_k$ such that e_is are (not necessarily distinct) members of $E(G)$ and v_is are (not necessarily distinct) members of $V(G)$, is called a walk in G provieded $\forall 1 \leq j < k \quad V(< e_j >) = \{v_j, v_{j+1}\}$. If $v_1 = v_k$, it is called closed walk. If all edges, appeared in its representations, are distinct, it is called trail. Moreover, a trail is called a path provided all vertices, appeared in its representation, are distinct. A cycle is a closed walk which in its representation at most one vertex is appears twice.

To dente a walk, e_is could be canceled; for example if the walk is in a simple graph. Moreover, in it, $v_1 v_2 \dots v_n$ and $v_1 W v_n$, provided the walk W be equal to $v_2 v_3 \dots v_{n-1}$, point the same walk.

Definition. Suppose a graph G. If there exists a collection of cycles $\mathcal{C}$ such that each edge of G is double covered by members of $\mathcal{C}$, the cover is called "cycle double cover". The existence of such this cover is denoted by $CDC(G)$. In cycle

double cover, using a member of $\mathcal{C}$ twice is allowed; for example a cycle must be applied twice to show $CDC(C_5)$.

It is accepted that the trivial graph with only one vertex v is planar which is covered by the trivial cycle: v!

Definition. A graph G is planar provided it has an embedding on the plane with no cross edges.

By the principle of explosion, the null graph which has any vertex is trivially planar.

Proposition 2.1. Every bridgeless planar graph has a cycle double cover.

Proof. Let G_i, $1 \leq i \leq k$, be a connectivity component of bridgeless planar graph G. Each G_i has an embedding on the plane with no cross edges such that each face is bounded by a cycle. (See for example theorem 10.7 of [1]). Let it has $R_i^1, R_i^2, \dots, R_i^{n_i}$ faces while infinite region be the face $R_i^{n_i}$. Denote C_i^j as the boundary of R_i^j. The collection of all cycle C_i^js, $1 \leq i \leq k$, $1 \leq j \leq n_i$, shows $CDC(G)$ obviously.

Definition. Two graphs G and H are called edge-disjoint if $E(G) \cap E(H) = \emptyset$. Moreover, they are called vertex-disjoint provided $V(G) \cap V(H) = \emptyset$.

Definition. Two paths tPu and vQw are called internally disjoint provided:

$$V(tPu - \{t, v\}) \cap V(vQw) = \emptyset \, , V(tPu) \cap V(vQw - \{v, w\}) = \emptyset.$$

Proposition 2.2. If for each $u, v \in V(G)$, there exists at least a pair of edge-disjoint path vPu and vQu, then G will be bridgeless.

Proof. If G has a cut edge $e = u_1v_1$, then between u_1 and v_1 there will be only one path. That is contradiction!

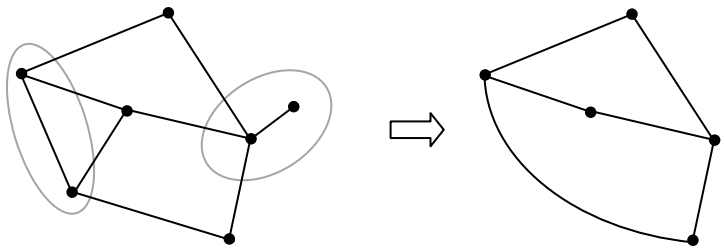
Proposition 2.3. If u, v, w are three different vertices of a bridgeless graph G, then there exist two paths vPu and vPw such that $V(vPu) \cap V(vQw) = \{v\}$.

Proof. In fact, the proposition is just rephrased, of a special case, of the fan lemma. To observe the fan lemma, for example, see proposition 9.5 of [1].

Proposition 2.4. If u and v are two distinct vertices of bridgeless graph G, then there exist two internally disjoint paths vPu and vQu.

Proof. For example, refer to theorem 5.1 of [1].

Definition. Let $A_1, \dots, A_k \subseteq V(G)$. The obtained underlined graph of identifying all members of each A_i will denoted by $\frac{G}{A_1,\dots,A_k}$. For example:



Clearly, if $A_i \cap A_j \neq \emptyset$, then all members of $A_i \cup A_j$ must will be identified in $\frac{G}{A_1,\dots,A_k}$. Moreover, note that if $A = \emptyset$ then $\frac{G}{A}$ is equal to G.

Notation. $\mathbb{K}$ denotes K_5 or $K_{3,3}$. $\mathbb{K}$ is usually called Kuratowski minor because of the next famous theorem.

Definition. Suppose a subdivision of graph H be a subgraph of G. H is called a topological minor of G; and it is written $H \preccurlyeq_T G$.

Theorem 2.5. (Kuratowski-Wagner) G is not planar if and only if $\mathbb{K} \preccurlyeq_T G$.

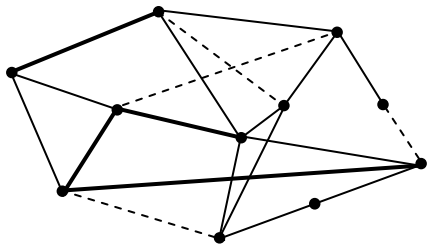
Proof. See, for example, theorem 4.4.6 of [2].

Naïve Cycle Double Cover

Definition. Suppose a graph G and its supergraph G'. An edge subset F of $E(G')$ is called a free edge set of G if

i. $\forall e \in F \quad |V(G) \cap V(< e >)| = 1$
ii. $\forall e, e' \in F \quad (V(< e >) \cap V(< e' >)) - V(G) = \emptyset$

A member of such F is called a free edge of G. Next picture shows a free edge set (dashed segments) of a graph (heavy segments).



Definition. If $v_1 \neq v_n$ and $P = v_1e_1v_2 \dots e_{n-1}v_n$ is a walk, then the edges e_1 and e_{n-1} are called terminal edges of P.

Definition. Let F be a free edge set of G. If there exists a collection of cycles and paths $\mathcal{C}$ such that each edge of $G \cup F$ is double covered by members of $\mathcal{C}$ while two terminal edges of each paths must cover two different free edges, the collection $\mathcal{C}$ is called a "naïve cycle double cover" of G and its free edge set F. The existence of such this cover is denoted by $ncdc(G;E)$. Moreover, $ncdc(G_1, \dots, G_n; F_1, \dots, F_k)$ means $ncdc(\cup G_i ; \cup F_i)$.

Notice that if $ncdc(G;F)$, then $|F| \neq 1$ because each covering path covers two distinct free edge; and moreover, G must be bridgeless.

Lemma 3.1. $ncdc(G;\emptyset) \leftrightarrow CDC(G)$.

Proof. Trival!

The next proposition can be used to reduce the study of general graph to simple graph.

Proposition 3.2. Suppose $\overline{G}$ be underlined graph of G.

i. $ncdc(\overline{G};F) \leftrightarrow ncdc(G,F)$,
ii. $CDC(\overline{G}) \leftrightarrow CDC(G)$.

Proof. Delete all loops from G to built $\ddot{G}$. After that, delete extra links to built $\bar{G}$.

Suppose the naïve cycle double cover $\mathcal{C}$ denotes $(\overline{G};F)$.

Choose $e_1 \in E(\overline{G})$ such that $V(< e_1 >) = \{u, v\}$. Therefore, there exist two members of $\mathcal{C}$ that they cover e_1. Without loss of generality, denote them by Pue_1vQ and $P'ue_2vQ'$. Let all edges e_1, e_2, …, e_k have the same ends u and v in G. Delete $P'ue_1vQ'$ from $\mathcal{C}$, and add the new $k+1$ paths:

$$ue_ive_{i+1}u, \qquad 1 \leq i \leq k$$

$$P'ue_kvQ'$$

The new cover, trivially, proves $ncdc(\overline{G}, < e_2, \ldots, e_k >; F)$. Repeat the procedure for each edge of $\overline{G}$ to shows $ncdc(\ddot{G}; F)$.

Now, choose $v \in V(G)$ that all loops e_1, e_2, …, e_k has just v as their ends. Suppose the naïve cycle double cover $\mathcal{C}$ denotes $ncdc(\ddot{G}; F)$.

Add the new k cycles $ve_iv, 1 \leq i \leq k$, to $\mathcal{C}$.

The new cover, trivially, prove $ncdc(\ddot{G}, < e_1, e_2, \ldots, e_k >; F)$ provided each new cycles are used twice. Repeat the procedure for each vertex of $\ddot{G}$ to show $ncdc(G; F)$.

This is complete the proof of one side. A similar structure can be used to prove the other side, and that will be completed the proof of "i". With similar methods, "ii" will be proved.

Lemma 3.3. Suppose G and H are vertex-disjoint graphs. If $ncdc(G; F)$ and $ncdc(H; F)$, then $ncdc(G, F, H; \emptyset)$.

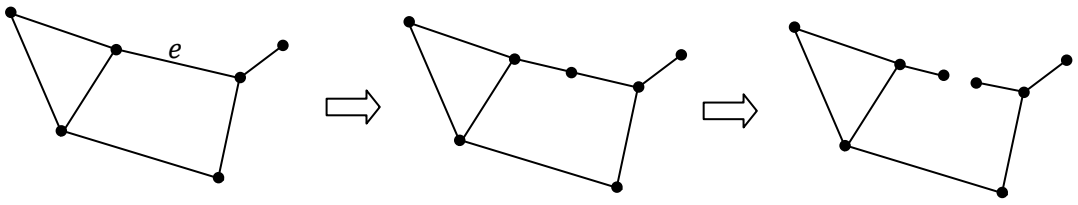
Proof. Suppose naïve cycle double covers $\mathcal{C}$ and $\mathcal{C}'$ denotes $ncdc(G; F)$ and $ncdc(H; F)$, in sequential order. For each $u_iv_i \in F$, there exists two paths Pu_iv_i and Qu_iv_i in $G \cup F$, and two paths $P'v_iu_i$ and $Q'v_iu_i$ in $H \cup F$. Delete these four paths of $\mathcal{C} \cup \mathcal{C}'$, and add the new paths Pu_iv_iP' and Qu_iv_iQ' instead of them. The constructed cover denotes $ncdc(G, F, H; \emptyset)$.

Corollary 3.4. Partition $V(G)$ to two vertex set X and Y.

$$ncdc(< X >; E(X, Y)), ncdc(< Y >; E(X, Y)) \rightarrow CDC(G)$$

Proof. Substitute $=< X >$, $H =< Y >$, and $F = E(X, Y)$ in the last lemma. After that use lemma 3.1.

Definition. Let $e \in E(G)$. Cut off e is done by acting two operations, in sequential order: subdivide e by adding a new vertex; and then split it on the new vertex. For example, in next diagram e is cut off.



Lemma 3.5. Suppose F be a free edge set of G, and $e = uu' \in E(G)$. Let cut off e to obtain two edges $f = uw$ and $f' = u'w'$. If $ncdc(G\backslash e; \{f, f'\}, F)$, then $ncdc(G; F)$.

Proof. Help to combine the suitable paths. It will have been trivial!

Definition. Let F be a free edge set of G. Define

$$cone(G; F) := \frac{G \cup < F >}{V(< F >) - V(G)}$$

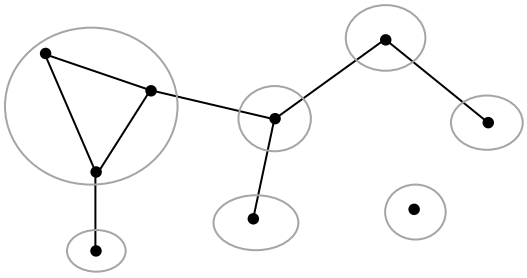
Note that $cone(G; \emptyset) = G$.

Proposition 3.6. $ncdc(G; F) \leftrightarrow CDC(cone(G; F))$.

Proof. Trivial! Just cut off each members of F.

Definition. A bridgeless component of G is defined as a maximal connected bridgeless subgraph of G.

Note that just one vertex might be a bridgeless component. For example, in next figure, G has seven bridgeless components.



In fact, each bridgeless components of a graph is its block. However, they have been named different because bridgeless component tree (look at the next definition) differs from block tree.

Definition. Suppose $G_1, G_2, \ldots, G_n$ are all bridgeless component of a connected graph G. Bridgeless component tree of G, denoted by T_G, is defined by $\frac{G}{<V(G_1)>,\ldots,<V(G_n)>}$. For each i, the shrunk vertex set $V(G_i)$ in T_G is denoted by v_{G_i}.

To be tree must be proved. In the block version, there is a proposition which implies:

Proposition 3.7. T_G is tree.

Proof. Trivial! For example, look at proposition 5.3 of [1].

Note that the definition could be generalized to disconnected graphs to introduce bridgeless component jungle.

Definition. A bridgeless component G_i of G is called terminal when $|\partial(G_i)| = 1$, and it is called isolated when $|\partial(G_i)| = 0$.

Lemma 3.8. Suppose G_1 and G_2 are two different bridgeless components of connected graph G.

i. If both of them are not terminal ones, there exist two terminal bridgeless components of G, named by G_1' and G_2', and three internally disjoint $v_{G_1'}Pv_{G_1}$, $v_{G_1}Qv_{G_2}$, and $v_{G_2}Rv_{G_1'}$ in T_G.
ii. If just one of them (for example G_1) is not terminal one, there exists a terminal bridgeless component of G, named by G_1' and two internally disjoint $v_{G_1'}Pv_{G_1}$ and $v_{G_1}Qv_{G_2}$ in T_G.
iii. If both of them are terminal ones, there exists a path $v_{G_1}Qv_{G_2}$ in T_G.

Proof. Consider tree T_G. The wanted results are just basic properties of to be tree.

Lemma 3.9. Suppose $v_1 \in V(G_1)$ and $v_2 \in V(G_2)$ while G_1 and G_2 are two different bridgeless components of connected graph G. Consider G_1 and G_2.

i. If both of them are not terminal ones, there exist three internally disjoint $v_1'Q_1v_1$, v_1Qv_2, and $v_2Q_2v_1'$ while v_i' is belong to terminal bridgeless component G_i', and $E(G_i') \cap E(< v_1'Q_1v_1Qv_2Q_2v_1' >) = \{v_i'\}$, for $i = 1,2$.
ii. If just one of them (for example G_1) is not terminal one, there exist two internally disjoint $v_1'Q_1v_1$ and v_1Qv_2 while v_1' is belong to terminal bridgeless component G_1', and $E(G_1') \cap E(< v_1'Q_1v_1Qv_2 >) = \{v_1'\}$.
iii. If both of them are terminal ones, there exists a path v_1Qv_2.

Proof. At first, zoom in item "i". Others can be proved in similar way, and to apply proposition 2.3.

The pervious lemma guarantees the existence of three internally disjoint $v_{G_1'}Pv_{G_1}$, $v_{G_1}Qv_{G_2}$, and $v_{G_2}Rv_{G_1'}$ in T_G. Define and label the set $\{H_1, H_2, \dots, H_r\} =$

$\{H|H$ is a bridgeless component of $G, v_H \in V(< P >) \cup V(< Q >) \cup V(< R >)\}$ such that H_i and H_{i+1} is joined by the unique edge, denoted by e_i^{i+1}. The uniqueness property is concluded, immediately, from the definition of bridgeless components.

Suppose $H_p = G_1,\ H_q = G_2$ while $p < q$. If it is impposible, just resort H_is reversely.

For $1 \leq i < r$, denote $e_i^{i+1} = u_i^{i+1} w_i^{i+1}$ such that $u_i^{i+1} \in V(H_i)$ and $e_i^{i+1} \in V(H_{i+1})$.

Each H_i $(i \neq p, q)$ is connected. Hence, there exists the path $w_i^{i+1} P_i u_i^{i+1}$ in H_i. Now, focus on H_p and H_q. In H_p, apply proposition 2.3 for at most three points $w_p^{p+1}, v_1, u_{p+1}^{p+2}$ to achieve the paths $w_p^{p+1} P_{v_1} v_1 Q_{v_1} u_{p+1}^{p+2}$. In the similar way, the path $w_q^{q+1} P_{v_2} v_2 Q_{v_2} u_{q+1}^{q+2}$ exists in H_q.

The next introduced path completes the proof.

$$u_1^2\ e_1^2 w_1^2 P_2 u_2^3\ e_2^3 w_2^3 \dots P_{p-1} w_p^{p+1} P_{v_1} v_1 Q_{v_1} u_{p+1}^{p+2} \dots P_{q-1} w_q^{q+1} P_{v_2} v_2 Q_{v_2} u_{q+1}^{q+2} \dots w_{r-2}^{r-1} P_{r-1} u_{r-1}^r\ e_{r-1}^r w_{r-1}^r$$

Definition. Suppose F be a free edge set of G. It is called surrounding free edge set if firstly, for each terminal bridgeless component G_i of G, $V(< F >) \cap V(G_i) \neq \emptyset$, and secondly, for each isolated bridgeless component G_i of G, $|V(< F >) \cap V(G_i)| \neq 1$.

Notation. Suppose F be a free edge set of G while $G_1, G_2, \dots, G_k$ be its connectivity components. Define $[G]_F \coloneqq \cup_{V(<F>) \cap V(G_i) \neq \emptyset} G_i$.

Proposition 3.10. If F is a surrounding free edge set of G, then $cone([G]_F; F)$ is bridgeless.

Proof. Choose $v_1, v_2 \in cone([G]_F; F)$ such that $v_1 \neq v_2$. It will have been shown the existence of two edge-disjoint paths $v_1 P v_2$ and $v_1 Q v_2$.

Firstly, suppose $v_1, v_2 \in V([G]_F)$. Let $v_1 \in H_1$ and $v_2 \in H_2$ such that H_1 and H_2 are bridgeless components of $[G]_F$. All five possible cases will be considered and studied.

Case 1-1. $H_1 = H_2$. There exists two internally disjoint paths $v_1 P v_2$ and $v_1 Q v_2$ by proposition 2.4.

All other cases include distinct H_1 and H_2.

Case 2-1. H_1 and H_2 are located in a same connectivity component H while both of them are non-terminal ones. Refer to the proof of lemma 3.9 to obtain the existence of the path $w_1^2Q_1v_1Qv_2Q_2w_{r-1}^r$ while w_1^2 and w_{r-1}^r are belong to terminal bridgeless components H_1', H_2' in sequential order. By definition of F, there exists a free edge w_it_i such that $w_i \in V(H_i')$, for $i = 1,2$. In addition, both of H_1' and H_2' are connected, so there exist paths $w_1R_1w_1^2$ in H_1' and $w_2R_2w_{r-1}^r$ in H_2'. Denote the identified vertices of t_1 and t_2 by t in $cone([G]_F; F)$. Trivially, there exist two internally disjoint paths v_1Qv_2 and $v_1Q_1w_1^2R_1w_1tw_2R_2w_{r-1}^rv_2$ in $cone([G]_F; F)$.

Case 3-1. H_1 and H_2 are located in a same connectivity component H while just H_1 is non-terminal one. Refer to the proof of lemma 3.9 to obtain the existence of the path $w_1^2Q_1v_1Qw_{r-1}^r$ while w_1^2 and w_{r-1}^r are belong to terminal bridgeless components H_1', H_2 in sequential order. On one By definition of F, there exists a free edge w_it_i such that $w_1 \in V(H_1')$ and $w_2 \in V(H_2)$. On one hand, H_1' is connected, so there exists a path $w_1R_1w_1^2$ in it. On the other hand, H_2 is bridgeless so by proposition 2.3, there exists a path $w_{r-1}^rR_2v_2R_3w_2$ in it. Denote the identified vertices of t_1 and t_2 by t in $cone([G]_F; F)$. Trivially, there exist two internally disjoint paths $v_1Qw_{r-1}^rR_2v_2$ and $v_1Q_1w_1^2R_1w_1tw_2R_3v_2$ in $cone([G]_F; F)$.

Case 4-1. H_1 and H_2 are located in a same connectivity component H while both of them are terminal ones. Refer to the proof of lemma 3.9 to obtain the existence of the path $w_1^2Qw_{r-1}^r$ while w_1^2 and w_{r-1}^r are belong to terminal bridgeless components H_1, H_2 in sequential order. By definition of F, there exists a free edge w_it_i such that $w_i \in V(H_i)$, for $i = 1,2$. In addition, both of H_1 and H_2 are bridgeless, so by proposition 2.3 there exist paths $w_1^2R_1v_1R_2w_1$ in H_1 and $w_{r-1}^rR_3v_2R_4w_2$ in H_2. Denote the identified vertices of t_1 and t_2 by t in $cone([G]_F; F)$. Trivially, there exist two internally disjoint paths $v_1R_1w_1^2Qw_{r-1}^rR_3v_2$ and $v_1R_2w_1tw_2R_4v_2$ in $cone([G]_F; F)$.

Case 5-1. $v_1 \in H_1$ and $v_2 \in H_2$ such that H_1 and H_2 are bridgeless components of G while contained in the different connectivity G_1 and G_2 of $[G]_F$, in sequential order. Cases 2,3-1, and case 1-2 will introduce two internally disjoint paths v_1S_1t and v_1T_1t and two internally disjoint paths v_2S_2t and v_2T_2t. Therefore, there are two edge-disjoint paths $v_1S_1tS_2v_2$ and $v_1T_1tT_2v_2$.

Secondly, suppose $v_1 \in V([G]_F)$, and $v_2 \notin V([G]_F)$. All two possible cases will be considered and studied.

Case 1-2. $v_1 \in V([G]_F)$ and $v_2 \notin V([G]_F)$ while v_1 is not a vertex of an isolated bridgeless component of $[G]_F$. Refer to cases 1,2,3-2, and change the name of vertices v_2 and t in their conclusions. According to argument, two internally disjoint paths will exist between v_1 and v_2.

Case 2-2. $v_1 \in V([G]_F)$ and $v_2 \notin V([G]_F)$ while v_1 is a vertex of an isolated bridgeless component vertex of $[G]_F$. By surrounding free edge set condition for isolated vertex, there are two path between v_1 and v_2.

Totally, these seven cases will show the existence of at least two edge-disjoint paths between each couple of distinct vertices. By proposition 2.2, the proof, now, have been completed.

The last proposition tells $\kappa'(cone([G]_F; F)) \geq 2$. (For instance, refer to the chapter 9 of [1], to catch the definition of $\kappa'(G)$ and $\kappa(G)$). There have been prepared a new result. It has no special role in this paper; however, it is:

Corollary 3.11. If F is a surrounding free edge set of a connected graph G, then $\kappa(cone(G; F)) \geq 2$.

Proof. Browse the last argument with the fact $[G]_F = G$. Notice that happening of "case 5-1" is impossible.

Proposition 3.12. If F is a surrounding free edge set of G, then $cone(G; F)$ is bridgeless.

Proof. Let $G_1, G_2, \dots, G_k$ be all connectivity components. Note that G_is are vertex-disjoint, and then

$$G = \left(\bigcup_{V(<F>)\cap V(G_i)\neq\emptyset} G_i\right) \cap \left(\bigcup_{V(<F>)\cap V(G_i)=\emptyset} G_i\right) = [G] \cap \left(\bigcup_{V(<F>)\cap V(G_i)=\emptyset} G_i\right).$$

Refer to the definition of $cone(G; F)$ when F is sourrounded free edge set to conclude $cone(G; F) = cone([G]_F; F) \cup \left(\bigcup_{V(<F>)\cap V(G_i)=\emptyset} G_i\right)$.

On one hand, F is surrounding free edge set of G, so for each i, if $V(< F >) \cap V(G_i) = \emptyset$, then G_i is bridgeless. On the other hand, proposition 3.10 guarantees $cone([G]_F; F)$ is bridgeless. Beside these two last conclusions, the proof has been completed.

Kuratowski Major

Definition. A closed walk W of graph G is called an especial walk if there exists an orientation of graph, denoted by $\overrightarrow{G}$, such that $\overrightarrow{W}$ in $\overrightarrow{G}$ covers each edge of $E(G)$ once or twice.

Formally, "a closed walk W is especial" means for each representation $\overrightarrow{W} = u_1e_1u_2 \dots e_{n-1}u_ne_nu_1$($u_i$s is not necessary distinct vertices) then

i. $\forall i,j \quad u_ie_iu_j \in W \rightarrow u_je_iu_i \notin W$,
ii. $\forall e \in E(G) \quad 1 \leq |\{i|e = e_i \in E(< W >)\}| \leq 2.$

Lemma 4.1. Suppose H is constructed by subdivision from G.

i. If G has an especial walk, then so do H.
ii. If edges of G are double covered by an especial walk W and collection of cycles, then there exist an especial walk and a collection of cycles that double cover the edges of H.

Proof. Trivial!

Proposition 4.2. Suppose edges of simple graph G are double covered by an especial walk W and collection of cycles $\mathcal{C}$. Let F be a free edge set of G such that $|F| \neq 1$. Then $ncdc(G;F)$.

Proof. Consider two cases:

Case1. $|F| > 1$.

Suppose $vw \in F$ such that $v \in V(G)$ and represent W by $u_1u_2 \dots u_{n(W)}u_1$ such that $u_1 \neq v$. If no this representation exists, then $|V(G)| = 1$ and it is easy to prove that $ncdc(G;F)$.

In the representation of W, v might appear several times in the string:

$$\begin{array}{cccccccccccc} u_1 & u_2 & \cdots & u_{i_1} & \cdots & u_{i_2} & \cdots & u_{i_{n(k_v)}} & \cdots & u_{n(W)} & u_1 \\ & & & \| & & \| & & \| & & & \\ & & & v & & v & & v & & & \end{array}$$

Choose just one of it; For example u_{i_l}. From W, introduce two walks:

$u_1u_2 \dots u_{i_l}w$, $wu_{i_l}u_{i_l+1} \dots u_nu_1$

Repeat the method for each free edge of G, again and again, to build walks ${}^1W$, ${}^2W, \dots,$ ${}^kW$ from W. Trivially the terminal edges of each ${}^iW$ are distinct free edges of G. (They are distinct because $|F| > 1$.)

Some ${}^iW$s might intersect themselves. Suppose for example ${}^iW$ intersect itself in the vertex v. Represent ${}^iW$ by $u'_1u'_2 \dots u'_{n({}^iW)}$ such that in $u'_{j_1} = u'_{j_2} = v$ while all vertices of the vertex string between u'_{j_1} and u'_{j_2} are equal neither to v nor to each other.

$$\begin{array}{ccccccccc} u'_1 & u'_2 & \cdots & u'_{j_1} & \cdots & u'_{j_2} & & u'_{n({}^iW)} \\ & & & \| & & \| & & \\ & & & v & & v & & \end{array}$$

Note that $j_2 - j_1 > 2$ because of the property of W:

$$\forall u, v \quad uv \in W \to vu \notin W \ .$$

Introduce two new walks from W:

$${}^iW^1 = u'_1u'_2 \dots u'_{j_1}u'_{j_2+1} \dots u'_{n({}^iW)} \qquad , \qquad {}^iW_1 = u'_{j_1}u'_{j_1+1} \dots u'_{j_2}$$

${}^iW_1$ is a cycle since, in the representation, all of its vertex are distinct expect its initial and terminal ones. In fact, there exists the orientation: $\{\overrightarrow{u_{J_1}u_{J_1+1}}, \overrightarrow{u_{J_1+1}u_{J_1+2}}, \dots, \overrightarrow{u_{J_2-1}u_{J_2}}\}$ which pass throw all its edges once because ${}^iW_1$, as a part of W, has the property:

$$\forall u, v \quad uv \in W \to vu \notin W \ .$$

For each i, repeat the method for the constructed ${}^iW^j, j \in \mathbb{N}$, again and again, till the cycles ${}^iW_1$, ${}^iW_2$, … are constructed. The process is finite because $|E({}^iW^1)| > |E({}^iW^2)| > \cdots$. For each i, denote the last member of the sequnse by ${}^iW^{p_i}$. The ends of the path ${}^iW^{p_i}$ are equal to the ends of the path ${}^iW$.

Define $\mathcal{D} := (\bigcup_{1 \le i \le k} \{ {}^iW_j \ | 1 \le j \le p_i \}) \cup \{ {}^iW^{p_i} \}$.

Obviously each edge of G which is covered by W once, is covered by members of $\mathcal{D}$ once; and each edge of G which is covered by W twice, is covered by members of $\mathcal{D}$ twice.

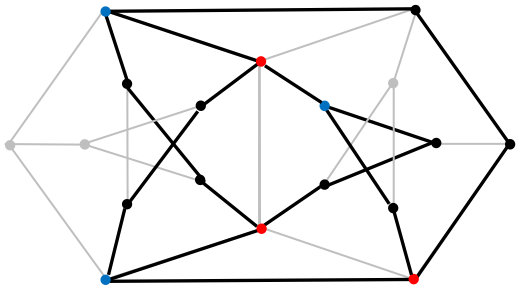
Since $\mathcal{C} \cup \{W\}$ double covers edges of G, then $\mathcal{C} \cup \mathcal{D}$ does it.

Case2. $|F| = 0$.

The argument is similar to the last case. Repeat the processes for the closed walk W, instead ${}^{i}W$s.

Definition. A subgraph H of graph G is called a Kuratowski major if H has been costructed by a subdivision of $\mathbb{K}$.

For example, in the next figure, a Kuratowski major is introduced. Six blue and red vertices correspond to six vertices of $K_{3,3}$.



Definition. Let H be a subgraph of G. Denote:

$$E_2(H) \coloneqq \{e \in E(G) - E(H)| \ |V(< e >) \cap V(H)| = 2\}$$

$$E_1(H) \coloneqq \{e \in E(G)| \ |V(< e >) \cap V(H)| = 1\}$$

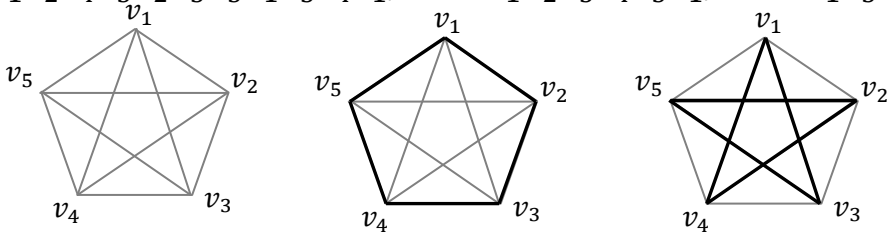
Proposition 4.3. Let F be a free edge set of Kuratowski major G. If $|F| \neq 1$, then $ncdc(G; F)$.

Proof. Firstly, let $\mathbb{K} = K_5$.

Check the next introduced closed walk $W_\circ$ is an especial walk for $\mathbb{K}$.

$W_\circ$ and the next cycles $C_\circ$ and $D_\circ$ introduce a double cover for $\mathbb{K}$. Therefore by lemma 4.1 and proposition 4.2 it will be proved $ncdc(G; F)$.

$W_\circ = v_1v_2v_4v_5v_2v_3v_5v_1v_3v_4v_1$, $C_\circ = v_1v_2v_3v_4v_5v_1$, $D_\circ = v_1v_3v_5v_2v_4v_1$.

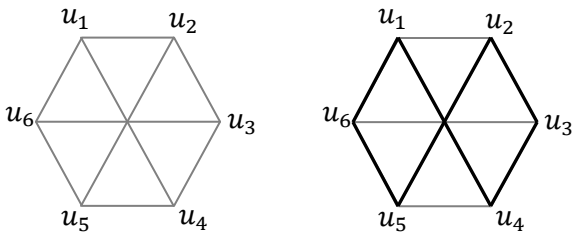


Secondly, let $\mathbb{K} = K_{3,3}$.

Check the next introduced closed walk $W_{\circ\circ}$ is an especial walk for $\mathbb{K}$.

$W_{\circ\circ}$ and the next cycle $C_{\circ\circ}$ introduce a double cover for $\mathbb{K}$. Therefore by lemma 4.1 and proposition 4.2 it is proved $ncdc(G;F)$.

$W_{\circ\circ} = u_1u_2u_5u_4u_3u_6u_5u_4u_1u_2u_3u_6u_1$, $C_{\circ\circ} = u_1u_4u_3u_2u_5u_6u_1$.



An Algorithmic Procedure

Consider the next algorithm.

input

G: a graph,

F: a surrounded free edge set of G

begin

$G_1 \coloneqq G; F_1 \coloneqq F; C_1 \coloneqq cone(G_1, F_1); i \coloneqq 1;$

while C_i is not planar then

$K_i \coloneqq$ a Kuratowski major of C_i;

$i \coloneqq i + 1;$

$G_i \coloneqq C_{i-1} - V(K_{i-1});$

$F_i \coloneqq E(V(G_i), V(K_{i-1}));$

$C_i \coloneqq cone(G_i, F_i);$

end.

The output of the algorithm is not unique because a Kuratowski major might be chosen differently for each G_i. For uniqueness, firstly, label all vertices of $G \cup F$ with distinct natural numbers. Name the labeling by L; secondly, associate a string (produced from vertex labeling) to vertex set of each subgraph; thirdly, use lexicographic order on the strings; fourthly, choose the Kuratowski major with minimum order; and fifthly, Repeat this method of labeling and choosing provided, in each identifying all members of F_i, the new vertex is labeled by $\prod_{j \text{ is the label number of } V(G_i)} j$. The output will be unique for each (G, F, L). The argument and proof of asserted facts guaranteeing the uniqueness are omitted because they are low importance topic in the main discuss.

Now, fix G and F under the conditions of the input of algorithm. All defined notations below are related to this fixed couple inputs.

Proposition 5.1. $\{i|C_i \text{ is produced by the algorithm}\}$ is finite set.

Proof. Note that $\forall i \ |V(K_i)| \geq |V(\mathbb{K})| \geq 5$. Hence $|V(C_{i+1})| \leq |V(C_i)| - 5$; and $|V(C_{i+1})| \leq |V(C_1)| - 5i$.

Therefore at most after i-time ($i > \left\lceil \frac{|V(C_1)|}{5} \right\rceil$) of running the "while" loop, the algorithm will have been finished.

Notation. $\mu \coloneqq max\{i|C_i \text{ is produced by the algorithm}\}$.

Corollary 5.2. C_μ is planar.

Proof. Trivial!

Proposition 5.3. If C_i is bridgeless and $i < \mu$, then F_{i+1} is a surrounding free edge set of G_{i+1}.

Proof. Suppose F_{i+1} is not a surrounding free edge set of G_{i+1}. Equally, just two cases are possible:

Case1. There exists a terminal bridgeless component A of G_{i+1} such that $V(A) \cap V(< F_{i+1} >) = \emptyset$. Denote the unique edge that connect A to the other component by e. It is obvious that the graph $A \cup < e >$ is a subgraph of C_i, while $\{e\} = E(V(A), V(C_i) - V(A))$ is a cut edge in C_i. It is contradiction with the bridgeless property of C_i!

Case2. There exists an isolated bridgeless component A of G_{i+1} such that $|V(A) \cap V(< F_{i+1} >)| = 1$. Denote the unique edge between A and $C_i - V(A)$ by e. Obviously, $\{e\} = E(V(A), V(C_i) - V(A))$. Hence, C_i has a cut edge while it was bridgeless. It is contradiction!

Proposition 5.4. $\forall 1 \leq i \leq \mu \quad C_i$ is bridgeless.

Proof. C_1 is bridgeless by proposition 3.5. Now, apply (and repeat) proposition 5.3 and 5.5 to conclude each of $C_2, C_3, \dots, C_\mu$ is bridgeless.

Proposition 5.5. $CDC(C_\mu)$.

Proof. By corollary 5.2, C_μ is planar. Moreover, it is bridgeless by proposition 5.2. $CDC(C_\mu)$ is achieved from proposition 2.1.

Proposition 5.6. $ncdc(K_i, E_2(K_i); E_1(K_i))$.

Proof. Compare $V(K_i)$ and $V(C_i)$. Just two cases are possible:

Case 1. If $V(K_i) = V(C_i)$, then $E_1(K_i) = \emptyset$. From proposition 4.3, $ncdc(K_i, ; \emptyset)$. Now apply proposition 4.3 with the technique introduced in lemma 3.5 for each member of $E_2(K_i)$ to get $ncdc(K_i, E_2(K_i); \emptyset)$ meaning $ncdc(K_i, E_2(K_i); E_1(K_i))$.

Case 2. If $V(K_i) \neq V(C_i)$, then $E_1(K_i) = E(V(G_{i+1}), V(K_i))$. By proposition 5.4, C_i is bridgeless, so $|E(V(G_{i+1}), V(K_i))| \neq 1$. Use proposition 4.3 to get $ncdc(K_i; E_1(K_i))$. Now apply proposition 4.3 with the technique introduced in lemma 3.5 for each member of $E_2(K_i)$ to prove $ncdc(K_i, E_2(K_i); E_1(K_i))$.

Proposition 5.7. $CDC(C_{i+1}) \rightarrow CDC(C_i)$.

Proof. In one hand, $ncdc(G_{i+1}; E_1(K_i))$ because

$$ncdc(G_{i+1}; E_1(K_i)) \xleftrightarrow{by\ definition} ncdc(G_{i+1}; F_{i+1}) \xleftrightarrow{by\ lemma\ 3.6} CDC(cone(G_{i+1}; F_{i+1})) \xleftrightarrow{by\ definition} CDC(C_{i+1})$$

On the other hand, $ncdc(K_i, E_2(K_i); E_1(K_i))$ by lemma G.

Now, combine two last results by corollary 3.3 to obtain $ncdc(G_{i+1}, K_i, E_2(K_i); \emptyset)$

Moreover,

$$ncdc(G_{i+1}, K_i, E_2(K_i); \emptyset) \xleftrightarrow{by\ definition} ncdc(C_i; \emptyset) \xleftrightarrow{by\ lemma\ 3.1} CDC(C_i)$$

Conclusions

Theorem 6.1. If F is a surrounding free edge set of graph G, then $ncdc(G;F)$.

Proof. Repeat prposition H to build the chain:

$$CDC(C_\mu) \to CDC(C_{\mu-1}) \to \cdots \to CDC(C_1)$$

Obviously, the first member of the chain, $CDC(C_\mu)$, is true by proposition F, so the last member of the chain, $CDC(C_1)$, will be true. Furthermore, $CDC(C_1) \xleftarrow{by\ definition} CDC(cone(G;F)) \xleftarrow{by\ lemma\ 3.6} ncdc(G;F)$.

Theorem 6.2. (Cycle Double Cover) Every bridgeless graph has a cycle double cover.

Proof. Denote a connectivity component of the graph by G. If G has only one vertex, G is planar, and the assertion is trivial. So suppose $|V(G)| > 1$.

G is bridgeless and connected. Moreover, $|V(G)| > 1$; so $|E(G)| > 1$. Choose an edge $e = uv$ of it. $G\backslash e$ has at most two bridgeless components that have u or v as a vertex. Cut off e, to obtain two free edge uu' and vv'. Define $F' := \{uu', vv'\}$. Trivially, F is a surrounding free edge set of $G\backslash e$. Now, from theorem 6.1, it is concluded $ncdc(G\backslash e; F')$. Moreover,

$$ncdc(G\backslash e; F') \xleftarrow{by\ lemma\ 3.5} ncdc(G\backslash e, F; \emptyset) \leftrightarrow ncdc(G; \emptyset) \xleftarrow{by\ lemma\ 3.1} CDC(G)$$

It means each connectivity component has a cycle double cover. This is complete the proof.

The theorem 6.1 prepares a strong tool to prove not only the cycle double cover problem (conjecture), but also a stronger version, conjectured by Goddyn:

Theorem 6.3. Fix a cycle C that is contained in a bridgeless graph G. There is a cycle double cover of G containing C.

Instead to prove it, the more generalized version will be proved:

Theorem 6.4. Fix edge-disjoint cycles C_i, $1 \le i \le k$, in a bridgeless graph G. There exists a cycle double cover of G containing each C_i.

Proof. Let $C \coloneqq \bigcup_{i=1}^{k} C_i$. Consider $E_2(C)$. Subdive each members of $E_2(C)$ to construct bridgeless graph G^*. Define $H := G^* - V(C)$, and $F := E(V(C), V(H))$. F is a surrounding free edge set of H. To prove it for contradiction, suppose F is not surrounding free edge set of H. Equally, there exists a bridgeless component A of H such that either A is an isolated one and $|V(A) \cap V(< F >)| = 1$, or A is a terminal one and $V(A) \cap V(< F >) = \emptyset$. In both case, there will exists a unique edge that connect A to the other component of G^*. Denote it by e. It means e will be a cut edge of bridgeless graph G^*; however, it is a contradiction.

Now, apply theorem 6.1 to prove $ncdc(H; F)$. Hence, there exists a naïve cycle double cover $\mathcal{C}$ such that each $e \in F$ is covered by two distinct members of $\mathcal{C}$. These paths are named by P_1^e and P_2^e. Define:

$$\mathcal{D} \coloneqq \{P_1^e, P_2^e | e \in F\}.$$

Denote all connectivity components of C by $D_1, D_2, \dots, D_p$. Trivially, each D_i , is a trail in G^*.

For $1 \le i \le p$, denote the trail D_i by $v_1^i v_2^i \dots v_{p_i}^i v_1^i$. Notice that some appeared vertices in the representation of D_i might equal. To define F_j^is, $1 \le j \le p_i$, decompose the set $\{e \in F | V(< e >) \cap V(D_i) \neq \emptyset\}$ to them, such that

i. $\bigcup_{j=1}^{p_i} F_j^i = \left\{e \in F \middle| v_j^i \in V(< e >)\right\}$
ii. $\forall 1 \le j < j' \le p_i \quad F_j^i \cap F_{j'}^i = \emptyset.$

To decompose, the greedy algorithm could be applied, or something else. Notice that some F_j^is might be empty set.

Denote each $F_j^i, 1 \le i \le p, 1 \le j \le p_i$, by $\{e_j^{i,1}, e_j^{i,2}, \dots, e_j^{i,q(F_j^i)}\}$. With pasting some paths, the set of $\mathcal{D}_i$ of new paths is constructed for each $1 \le i \le p$.

$$P_2^{e_j^{i,r}} v_j^i P_1^{e_j^{i,r+1}} \quad ,1 \le j \le p_i \ ,1 \le r < q(F_j^i)$$

$$P_2^{e_j^{i,q(F_j^i)}} \underbrace{v_j^i v_{j+1}^i \dots v_{s(i,j)}^i}_{\text{a path in } D_i} P_1^{e_{s(i,j)}^{i,1}} \text{ if } F_j^i \neq \emptyset, s(i,j) = \min\{l | F_l^i \neq \emptyset, j < l', l \equiv l' (mod\ n)\}$$

It can be seen that $\mathcal{C}' \coloneqq (\mathcal{C} - \mathcal{D}) \cup (\bigcup_{i=1}^{p}(\mathcal{D}_i \cup \{D_i\}))$ double covers each edge of G^*. Moreover, instead of $\bigcup_{i=1}^{p}\{D_i\}$, $\bigcup_{i=1}^{k}\{C_i\}$ can be replaced. Therefore, $\mathcal{C}'' \coloneqq (\mathcal{C} - \mathcal{D}) \cup \left(\bigcup_{i=1}^{p}\mathcal{D}_i\right) \cup (\bigcup_{i=1}^{k}\{C_i\})$ double covers each edge of G^*. It means $\mathcal{C}''$ which shows $CDC(G^*)$ includes edge-disjoint cycles $C_1, C_2, \dots, C_k$. Immediately, there will be existed a cycle double cover $\mathcal{C}'''$ of G, including edge-disjoint cycles $C_1, C_2, \dots, C_k$. It is complete the proof.

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